



Improved 3-D Protein Structure Predictions using Deep ResNet Model

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Accepted: 9 August 2021 / Published online: 12 September 2021

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Abstract

Protein Structure Prediction (PSP) is considered to be a complicated problem in computational biology. In spite of, the remarkable progress made by the co-evolution-based method in PSP, it is still a challenging and unresolved problem. Recently, along with co-evolutionary relationships, deep learning approaches have been introduced in PSP that lead to significant progress. In this paper a novel methodology using deep ResNet architecture for predicting inter-residue distance and dihedral angles is proposed, that aims to generate 125 homologous sequences in an average from a set of customized sequence database. These sequences are used to generate input features. As an outcome of neural networks, a pool of structures is generated from which the lowest potential structure is chosen as the final predicted 3-D protein structure. The proposed method is trained using 6521 protein sequences extracted from Protein Data Bank (PDB). For testing 48 protein sequences whose residue length is less than 400 residues are chosen from the 13th Critical Assessment of protein Structure Prediction (CASP 13) dataset are used. The model is compared with Alphafold, Zhang, and RaptorX. The template modeling (TM) score is used to evaluate the accuracy of the estimated structure. The proposed method produces better performances for 52% of the target sequences while that of Alphafold, Zhang, RaptorX were 10%, 22.9%, and 6% respectively. Additionally, for 37.5% target sequences, the proposed method was able to achieve accuracy greater than or equal to 0.80. The TM score obtained for the sequences under consideration were 0.69, 0.67, 0.65, and 0.58 respectively for the proposed method, Alphafold, Zhang, and RaptorX.

Keywords Protein · 3-D protein structure prediction · CASP · Experimental and Computational techniques · Deep ResNet Architecture · Distance prediction

1 Introduction

Deoxyribonucleic acid (DNA) contains the code to generate protein sequence which is found inside the nucleus of the cell. A DNA molecule undergoes transcription and translation process to form a protein sequence this process is known as the protein formation. The transcription is a process that takes place inside nucleus of the cell where DNA molecules transfer the code to mRNA. The mRNA carries this code outside the nucleus where translation takes place. The translation is a process by which protein sequence is

formed from mRNA. Any mutation that occurs during the process of protein formation can make change in protein folding and may result in a negative or positive impact on the organism. For instance, any alternation in DNA coding may lead to either enhancement of health or onset of disease in a living organism. Protein is a tangled biological molecule that plays a salient part in assembling, fortifying, keeping up, securing, and fixing a creature. Every protein bends into a distinct 3-D diagram depending upon its unique amino acid arrangement [1].

The unique folding of protein helps it to do different functions in a living organism [2]. For the last few decades, the researchers and scientists are trying to figure out how this protein folding takes place is known as protein folding problem. Researchers are experimenting to sort out this problem by applying various techniques. Once we know about how proteins fold we get more thorough understanding of the protein molecules and their association with diseases and how they function. Thus, researchers can produce new drugs with

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limited resources that work well with the unique structure of the protein [1].

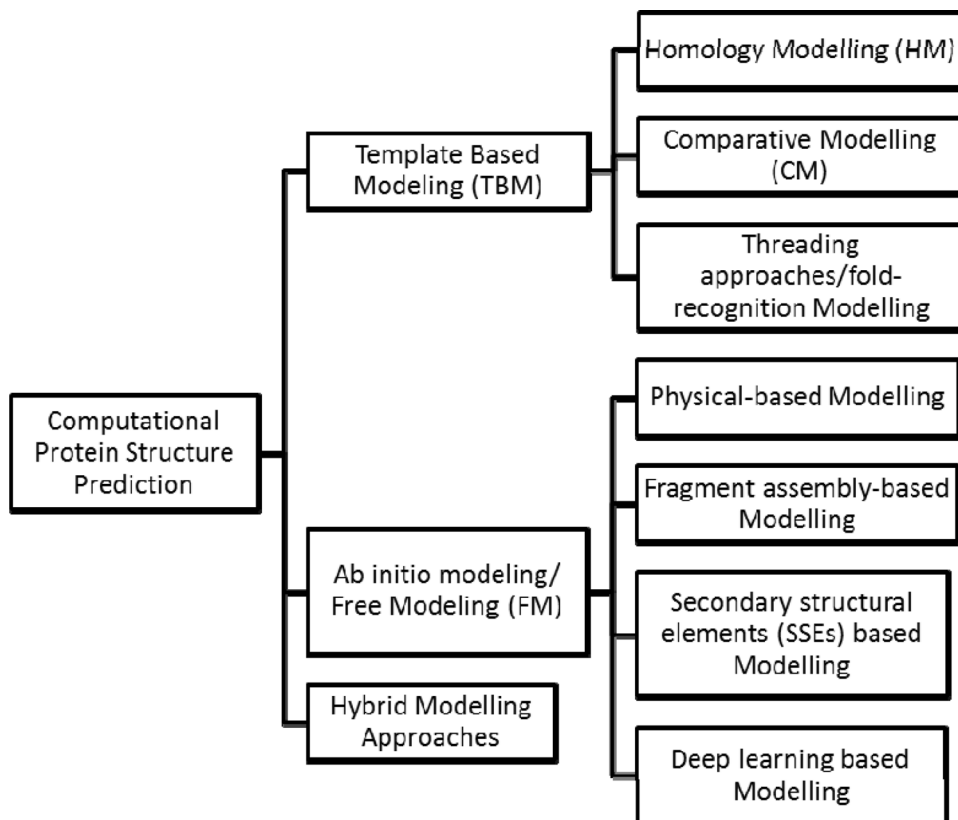
The experimental methods used for protein structure prediction (PSP) are nuclear magnetic resonance spectroscopy (NMRs) [3], X-ray crystallography, and Cryo-EM. NMRs gives a lower resolution of protein structure which is limited to modest proteins and deeper information about conformational changes of a protein. X-ray crystallography provides a high-resolution 3-D shape of protein, yet it doesn't give data about conformational changes of a protein. It can be applied to both small and long proteins. Cryo-EM [4] has recently developed experimental methods that have encouraged a speeding up of protein structure assurance measures. The basis of this technique is low resolution design using frozen photograph pieces. Even though these methods are proven to be the best effect, they are time-consuming and costly, and need an expert in the field to conduct the experiment which consistently increased the space between a protein and its structural layout. The use of bioinformatics in this issue helps us to develop new drugs within a limited time and with minimum side effects which are rarely harmful than the experimental approach. This lead to the evolution of computational procedures for PSP.

CASP [5] is a biennial competition that has been taking place every two years since 1994. It has developed over time into a computational PSP system that certifies the latest

methodologies. CASP organizes a collection of unpredicted protein sequences for each discussion in a regular interval whose protein shape is yet to be disclosed to assess progress in PSP techniques. The models are evaluated based on several metrics like the template modeling (TM) score [6] and the GDT_TS score [7]. The underlying accomplishment in PSP was seen in 4th CASP, but only for protein sequences less than or equal to 120 residues. The further update took place in CASP 11 with the launch of advanced contact estimation procedures. It enhanced a promising level of accuracy in three-dimensional PSP. In the 13th CASP deep learning approach is utilized to deal with the PSP issue, which brought an unbelievable improvement in this area [8]. CASP has classified the computational technique of PSP generally into three categories as template-based modelling (TBM) [9], Ab initio modelling/free-modelling (FM) [10], and Hybrid approaches. Different classifications of computational methods are illustrated in Fig. 1.

Template-based modelling [11] utilizes the template to estimate the three-dimensional protein structure. Based on sequence alignment similarity, it can be sub-divided into homology modelling (HM): The homology modeling approach is used when there exists an evolutionary relationship with the target sequence and have a high sequence identity. Comparative modelling (CM) is applied when there is no ancestral relationship with the target sequence

Fig. 1 An overview of different computational methods



but has a high sequence identity. Threading approaches/fold-recognition follows, picking layout structures depending on their suitable query sequence. It is an association between the linear target sequence and the three-dimensional protein structure.

Ab initio modeling/free-modeling (FM) *Ab initio* modeling is also known as free modeling. Many sequences do not have homology sequence similarity (ancestry similarity) with the existing set of sequence databases. In that case, we go for this modeling for the PSP which estimate 3-D protein structure by considering either minimal or unknown information. It uses experimental data [12, 13] so they do not require a template. Based on differences in the use of methodology this approach can be further subdivided into: In physical-based method, molecular dynamics (MD) simulation is employed to bind the protein. For instance, modeling employing limited data (MELD) [14] and D. E. Shaw Research group [15] are the major development under this approach. The fragment assembly-based method was introduced by Bowie and Eisenberg [16]. This method uses the fragment of known protein and an energy function to predict the stable structure. This approach uses the fragment length, type, and scoring functions to produce enthusiastically limited stable structures. The well-known approaches which contribute much to this method are Rosetta [17] and QUARK [18]. Secondary structural elements (SSEs) based approaches utilizes SSEs and mainly concentrate on getting together the basics of the protein. BCL: Fold [19] and SSThread [20] are the main contributors to this method. Deep learning-based approaches use the neural network for training. It make use of either a co-evolutionary relationship or template to predict the three-dimensional structure. Recurrent geometric network (RGN) [21] and AlphaFold [8] are the two main prominent methods that come under these approaches.

Hybrid approaches can be an integration of one or more methods of either TBM or FM or both. The major works in this approach are Bhageerath [22] which is a composition of both homology modeling, which comes under the TBM approach, and FM approach. Later, it is upgraded as Bhageerath H⁺ [23] and tested successfully on CASP10 and CASP12. I-Tasser pipeline [24, 25] that utilizes both QUARK [18] and fragment-guided molecular dynamic (FG-MD) method generated better results in CASP 10 AND CASP 11 FM target.

The first method RGN [13, 21] which incorporated the pure deep learning approach for PSP was proposed by Al Quraishi. This method uses a position-specific scoring matrix (PSSM) and only allows one residue at a time, along with the query sequence. Still, it can produce better results in the free modeling approach. The second method that brought a major impact on this approach is AlphaFold, which made use of co-evolutionary relationships in PSP.

Though there can be other methods that make use of deep learning approaches for PSP. This paper focuses on those methods that utilized neural networks in their work, and also used the CASP13 dataset as their test set. Deep ResNet architecture with dilation is used along with co-evolutionary relationships to predict distance and direct optimization is used to generate 3-D structure in RaptorX [26], Zhang [27] and AlphaFold [21].

2 Materials and Methods

This paper concentrates on the deep learning approach employed for the PSP. In this work, a deep ResNet networks are utilized to predict inter-residue distance and dihedral angles. A query sequence is given to a customised database (Uniclust30 (2017–10), UniRef90 (2018–04), and Uniref100 (2018–04), metagenomics sequence database (2018–04), and NR90 database (2016)) that generate MSA. The features are extracted from MSA. These features are given as input to two different ResNet architectures to predict inter-residue distance and dihedral angles respectively. These predicted values are used to compute 3-D co-ordinate values. Stochastic gradient descent is used to optimize the resulting potentials to generate a 3-D protein structure. A pool of structure is generated from which the lowest mean potential 3-D structure is chosen as the final structure. The proposed method's flowchart is shown in Fig. 2.

These are the steps followed by deep ResNet architectures to predict 3-D protein structure:

- (1) Retrieval of similar sequences according to the query sequence from the databases.
- (2) Feature extraction is carried out based on the above sequences.
- (3) Training of Deep ResNet using 6521 protein sequences from the Deepcov method.
- (4) For testing the network 48 protein sequences of residue length less than 400 residues from the CASP13 dataset are used.
- (5) Deep ResNet architectures are utilized to estimate dihedral angles and inter-residue distance.
- (6) Finally, 3-D coordinates are generated to reconstruct the corresponding protein structure.

2.1 Multiple Sequence Alignment (MSA) Generation

MSA is used for identifying the structural, functional, and evolutionary relationship between protein sequences. Metaclust50 (2018–01) (a union of metagenomics sequence database (2018–04) [28] UniRef100 (2018–04), and NR90 database (2016)), UniRef90 (2018–04) [29], and Uniclust30 (2017–10) [30] are protein databases used to generate MSA

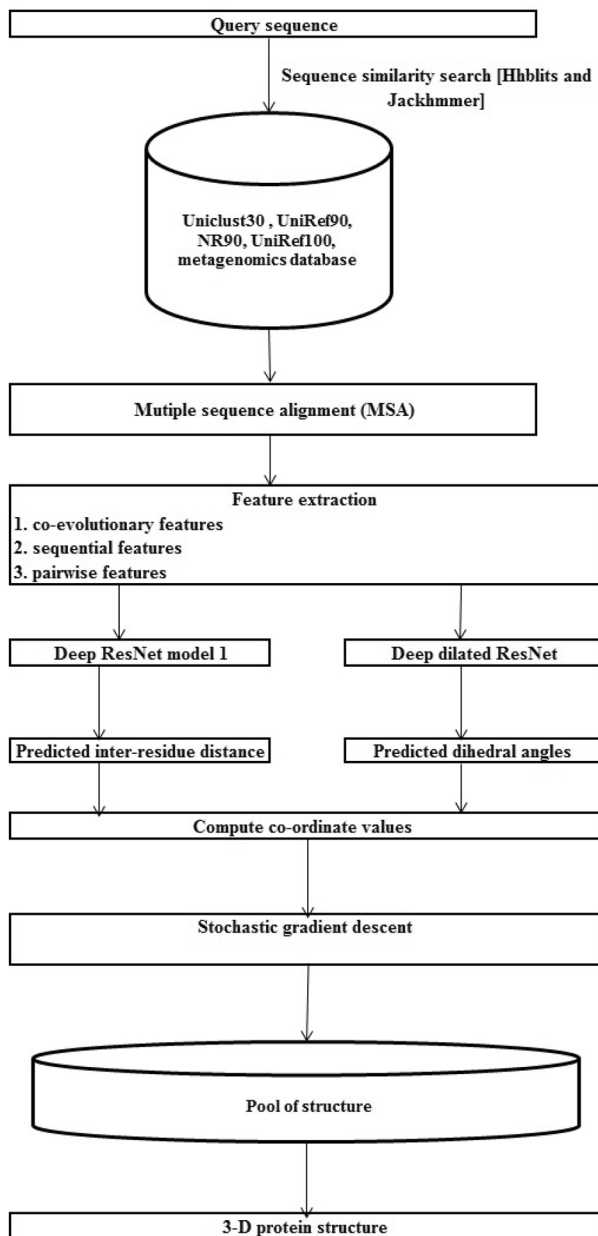


Fig. 2 The workflow of the proposed method

alignment. By examining homologous sequences, it is possible to conclude contact between amino acid residues this helps in the prediction of protein structures [31]. MSAs are crucial for distance/ contact prediction based on co-evolutionary analysis. DeepMSA approach is used to generate MSA process.

From the previous studies it can be concluded that DeepMSA can improve the accuracy of prediction [32, 33]. To check the homologous of the multiple sequences alignment obtained through this process N_f [34] (normalized no: of effective sequence) is used. To attain accurate contact

prediction the N_f is set as 128. The general formula for calculating N_f is given below:

$$N_f = \frac{1}{\sqrt{L}} \sum_{n=1}^N \frac{1}{1 + \sum_{m=1}^N \pi[S_n, m \geq 0.8]} \quad (1)$$

where n and m are sequences

if sequence identity between m and n

$$\begin{cases} > 0.8, & \pi[S_n, m \geq 0.8] = 1 \\ \text{otherwise,} & \pi[S_n, m \geq 0.8] = 0 \end{cases}$$

The process to generate MSA is as follows. In first stage, by setting minimal coverage as 50% searching is done against Uniclust30 (2017–10) database for three iterations using HHblits [35]. If N_f is less than 128 in first stage then the process continues with second stage. Otherwise, the first stage MSA is set as the final MSA. In second stage UniRef90 (2018–04) searching is done using Jackhmmer [36] for three iterations to generate MSA with E-value as 10. “hhblitdb.pl” script from HH-suite for making a custom database from the Jackhmmer hits for searching for more HHblits. The third stage is performed if N_f is still lower than 128 obtained in second stage. HMMbuild from the HMMER package searches against a customized database which consists of Uniref100 (2018–04), metagenomics sequence database (2018–04), and NR90 database (2016). A custom HHblits database is built from the hits, similar to second stage. The second stage and third stage contains MSA generated from previous stages by applying HHblits searching. Compared with the default programs used in other methods the accuracy of long-range contacts increased up to 24.4% and also 7.5% increases have been observed in the average TM-score of the template alignments. This method is able to produce result for targets without homologous sequences in the database.

2.2 Input Features Generation

The main features considered for predicting inter-residue distance and dihedral angles are co-variance matrix, pseudo-likelihood maximization, precision matrix, and, Karl’s Pearson correlation (r), protein specific scoring matrix (PSSM), solvent accessibility, secondary structure, Shannon entropy (SE), mean contact potential, normalized mutual information (NMI), contact-score, mutual information (MI).

The co-evolutionary features include co-variance matrix, precision matrix, and pseudolikelihood maximization. The covariance matrix encrypts the correlations between variables Suppose MSA as $M \times N$ matrix where M is the number of columns and N is the number of rows. We can compute

the covariance matrix of dimension $21L \times 21L$ (20 amino acids and 1 gap penalty) as:

$$Cov_{ij}(x, y) = R_{ij}(x, y) - R_i(x)R_j(y) \quad (2)$$

where $R_{ij}(x, y)$ is the observed corresponding frequency at location i and j for the pair of residue x and y ; $R_i(x)$ and $R_j(y)$ frequency of occurrence of a particular residue x and y at locations i and j respectively.

Each element in the covariance matrix indicates covariance of residue x at location i with residue y at location j . Equation two is the general equation used for calculating the covariance matrix. There are cases where the covariance matrix fails to produce a better result. The precision matrix and pseudolikelihood maximization are also included to overcome the failure that occurs due to the covariance matrix. The precision matrix is the negative loglikelihood of the inverse of the covariance matrix. The precision matrix and pseudolikelihood maximization are calculated as used in TripletRes and resTriplet [37]. It provides information regarding direct coupling which is used to improve the performance of contact prediction. Pseudolikelihood maximization is considered to approximate the inverse of the Potts model. The Potts model is used to compute the intensity of interconnection between amino acid residues.

The sequential features consist of solvent accessibility, sequence profile (PSSM), and secondary structure. The features such as solvent accessibility and sequence profile (PSSM) are generated from PSIPRED [38] and PSI-BLAST [39] respectively. Prediction of secondary structure is done by using dictionary of secondary structure of proteins (DSSP) classification.

Normalized mutual information, SE sum, mutual information, and mean contact potential, contact scores are the features that are included in pairwise features [40]. Contact scores are generated using CCMpred [41]. Pearson's correlation is also calculated. By averaging contact potential terms across the two alignment columns mean contact potential is computed. The C program in MetaP-SICOV [42] is used to compute mean potential and SE at each residue position. MI is the most useful for proteins with a large number of sequence homologs, as it is close to 0 when only a few exist. NMI is a scaling of the MI score between 0 and 1. Zero and One in NMI stand for no mutual information and complete mutual information respectively. This metric isn't adjusted for randomness. SE of each sequence is determined, very high SE indicates that the protein was disordered and had a significant impact on illness causes.

There are some cases where evolutionary information alone cannot produce the result. The non-evolutionary pieces of information are also included that improved the

accuracy [26]. All these features are used to train and predict inter-residue distance as well as dihedral angles.

2.3 The Proposed Architecture

Recent work in PSP shows that the prediction accuracy has improved with the use of co-evolutionary relationships and the deep learning approach. Though contact prediction methods are widely in use, many methods use distance prediction. Since distance prediction can decrease conformational sampling to a great extent. It can provide more information and also avoid inconsistency between actual and estimated contact probability. So, this model makes use of distance prediction (multi-class classification) instead of contact prediction (binary classification). The Fig. 3 illustrates the proposed method's overall structure.

In this study, ResNet architecture, a type of convolutional neural network (CNN) that has contributed towards the progress in PSP is considered. A protein sequence is taken as input to the proposed method. From Fig. 2, it can be inferred that the input features are extracted before submitting it to the neural network. The deep ResNet architectures are used to estimate dihedral angles and inter-residue distance. Then, from these predicted values protein 3-D structure is plotted. Here a detailed explanation regarding each ResNet model used for predicting inter-residue distance and dihedral angles are given below. The Supplementary Fig. 1 shows the workflow of the inter-residue distance.

First deep ResNet architectures that used to predict the inter-residue distance are explained. The distance between

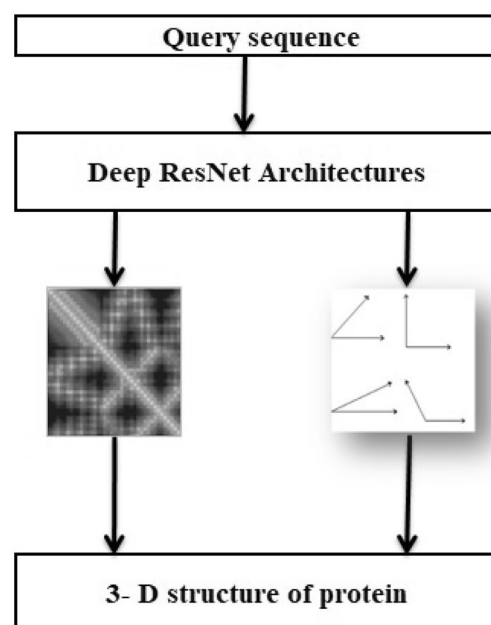


Fig. 3 The overall structure of proposed method

residues of pairwise beta carbon (C_β) atoms (alpha carbon (C_α) of glycine a special case) was classified into 26 classes. These 26 classes are subdivided into six classes. One class each for distances less than $4.5 \text{ \AA}$, greater than $20.0 \text{ \AA}$, and gap or missing position. The rest 23 classes with distance vary between $[4.5-6 \text{ \AA}]$, $[6-12 \text{ \AA}]$, and $[12-20 \text{ \AA}]$ are used with varying class intervals as 0.5, 0.6, 0.8 respectively. For predicting inter-residue distance input features are grouped. Each set includes a mix of both co-evolutionary and non-evolutionary features. The first set consists of Precision matrix, contact scores, PSSM, and Pearson correlation. The second includes Pseudolikelihood maximization, PSSM, and Pearson correlation. The third set contains covariance matrix, SE sum, PSSM, and Pearson correlation. And the fourth set includes solvent accessibility, SE sum, mutual information, mean contact potential, and normalized mutual information. Figure 5a is the overall structure for predicting inter-residue prediction. Each set to four different ResNet architecture.

Normalization is employed to the input features to speed up the learning process. As the first steps, Column Normalization, Row Normalization and Instance Normalization (RCIN) is applied to the input features. The application of all these normalizations ensures the contribution of every feature and reduces the risk of exploding gradient. After normalization rectified linear unit (ReLU) activation function is applied. It is followed by one convolution layer of size 3×3 , which creates feature maps that summarize the presence of all input features. Since a small movement in the position of the feature in the feature map can create a different feature map. That leads to the development of the different outcomes. To avoid this situation and to reduce the computational cost and complexity downsampling is done. Max pooling is used for downsampling that reduces the input feature to 64. The residual block receives the resultant as an input [34].

The structure of the residual block is given in Fig. 4 below. Each residual block consists of two RCIN, two ReLU activation functions, two convolution layers of size 3×3 , and SE_block [43] at the end of the residual block. The internal convolution layer extracts more abstract and in-depth information. Altogether sixty such residual blocks are used. The instance normalization layer is added to the last residual block and then the ReLU activation layer is applied. The softmax function is used after the ReLU activation layer to estimate the multi-class classification with 26 output classes. The average of the result sets is considered as the final inter-residue distance map.

The estimated distances about the diagonal line are symmetrical. Those values which are far away from the diagonal line (long-range) are very crucial to improve the accuracy of the prediction [44]. The traditional method of calculating contact prediction from distance, such as distances smaller

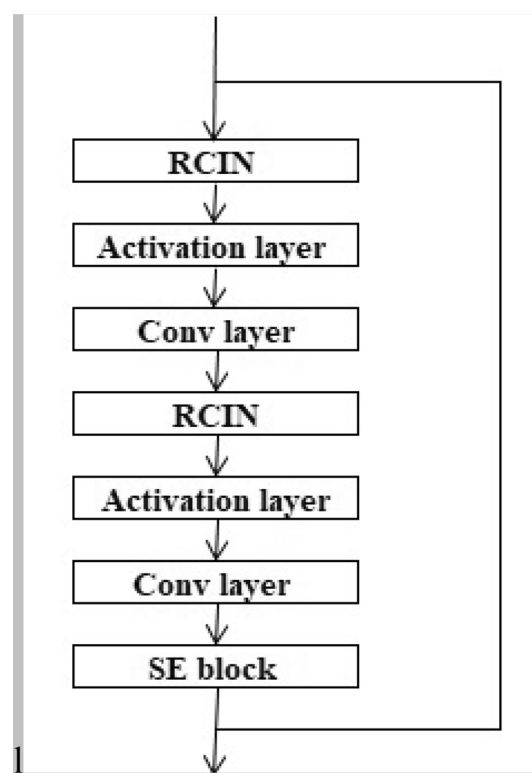


Fig. 4 Residual block

than $8 \text{ \AA}$ as contacts, does not favour distance prediction evaluation [45]. So to evaluate the estimation of distance prediction the similar method used in this paper [46] is used. The formula for calculating C_{ij} is given below.

$$C_{ij} = \begin{cases} 0.5, & \text{if } de = 8.0 \\ \frac{4.0}{de}, & \text{if } de \geq 4.0 \\ 1, & \text{otherwise} \end{cases} \quad (3)$$

where 'de' stands the estimated distance between i th and j th residue of the protein sequence.

In general, there are three angles which are considered as torsion angles (φ , ψ , ω). Among these three angles, the variation in ω is negligible thus ψ and φ angles are used. The angles (ψ , φ) of the 3-D structure is classified into 37 classes that include one for gaps and -180° to 180° with a class width of 10. To obtain backbone angles (φ , ψ) of protein structure, we trained a separate neural network that is executed parallel to the distance prediction network. Similar to distance prediction this network accepts an amino acid as input and passes it to database collections to generate MSA. MSA is used to generate all input features.

Figure 5b is the overall structure for predicting dihedral angles. 2 D ResNet architecture is used for predicting dihedral angles for which all input features are accepted. First,

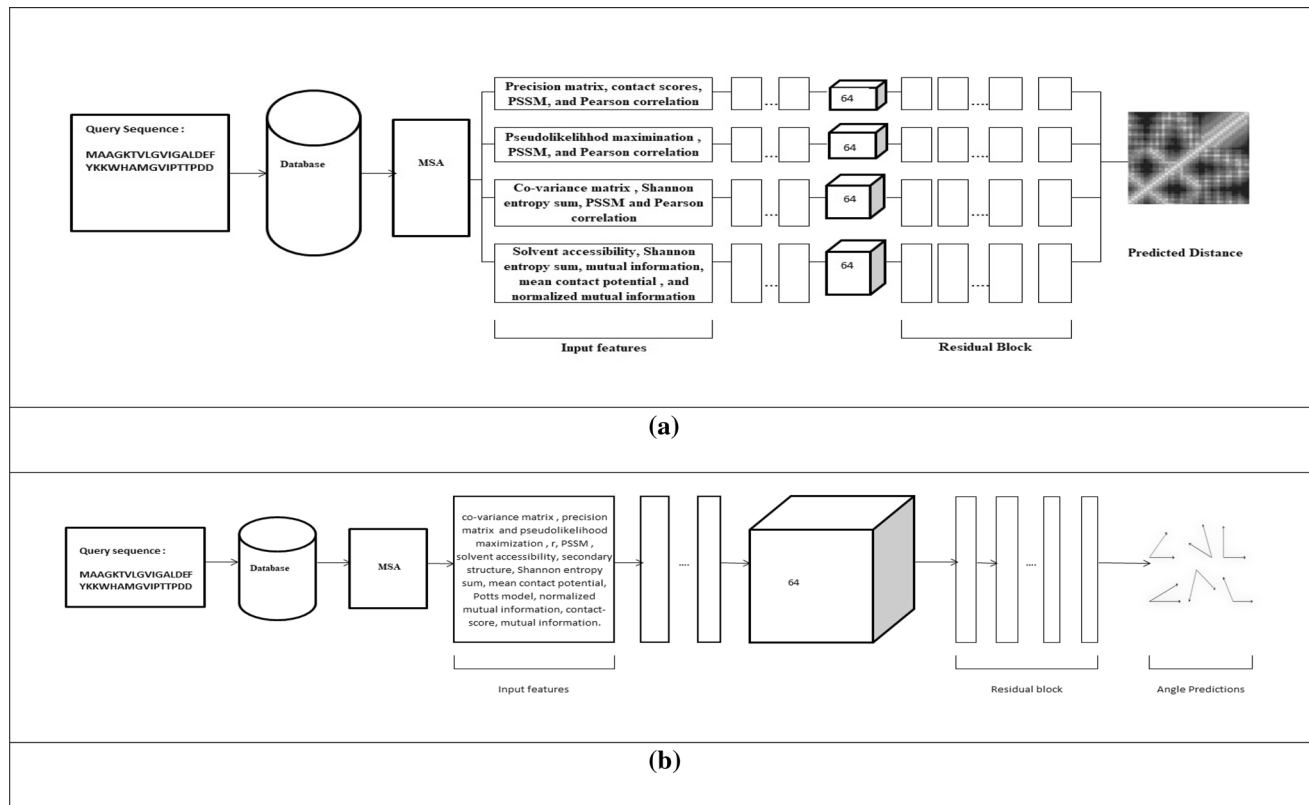


Fig. 5 **a** An pipeline of distance prediction and **b** an pipeline of dihedral angles prediction

RCIN are applied to the input features. It is followed by the ReLU function, and one convolution layer. Each residual block is made up of the layers instance normalization layer, ReLU activation function, convolution layer, batch normalisation (batch_norm), ReLU activation function, dilated convolution layer, instance normalization layer, and upsampling in the sequence provided. The right hand side of the Supplementary Fig. 1 shows the workflow of the dihedral angles. The batch normalization is used to normalize the contributions to a layer for every mini-batch. Each convolution output contains a wide range of information. To retain dimensionality dilated filter size of 1, 2, 4, and 8 strides are used. Sixty such residual blocks are used. The ReLU function, then one convolution layer, follows the last residual block. At last softmax activation layer is applied to classify dihedral angles.

The discrete probability distribution is applied over the predicted values are integrated to form coordinates of 3-D structure. This part of model is similar to that of Alpha-fold. All the resultant probabilities are normalized by the corresponding probability and smoothing by applying the spline function in Rosetta. Von-Moses distribution and Vander walls term is used to smooth angles and avoid steric clashes respectively. By setting a maximum threshold value to hundred a collection of protein structures is obtained. The

protein structure with the lowest mean potential is chosen as the final 3-D structure from the pool of structures. To optimize scores a mathematical technique known as stochastic gradient descent is used that result in a highly accurate 3-D structure of a protein.

2.4 Training Phase

The training dataset consists of 6521 protein sequences, a subset of training datasets used in the Deepcov method [45]. We downloaded structural data from the Protein Data Bank (PDB) for each FASTA file in the Deepcov method. It is cleaned to match with the corresponding FASTA file by removing unwanted non-standard amino acids. The DSSP database contains a secondary structure for each entry in the PDB. The secondary structure is classified into 9 classes includes one of the missing position (gap) along with 8 classes of DSSP. The angles of the 3-D protein structure are phi, psi, and omega (φ , ψ , ω). The variation in ω is negligibly small so this angle is not used for 3-D construction.

The proposed method is validated using 208 protein sequences from the DeepCov method. Since sequence having sequence-identity less than 30% can belong to the same family and with homogenous structure there is a chance

to copy the same without making the prediction. To avoid this situation we applied ECOD database classification [47] which makes use of evolutionary relationships to classify the protein domain hierarchically. If a protein sequence was in the same ECOD [48] T-group as training set, it was removed from the validation set. The protein sequences used in both the training and validation dataset are included before the CASP 13 experiment (Supplementary excel file).

2.5 Testing Phase

For testing, we had used 48 protein sequences from the CASP 13 dataset where residue length is less than 400 residues. These protein sequences are publicly available on the CASP website under the target list. The Seq_Id and Seq_len correspond to Tar_id and Res in CASP website respectively. Res stand for the number of residues in the corresponding protein sequence. The Seq_Id is a unique id to represent the target sequence address as in the CASP 13 dataset. The length of the relevant protein sequence is represented by Seq_len. Table 1 of supplementary file is the list of 48 protein sequences used for testing the model.

3 Results

Deep ResNet has been utilized by different groups in the CASP13 experiment. But, here we are considering only those top global methods in CASP13 that make use of ResNet in their work like AlphaFold, Zhang, and RaptorX along with the proposed method. Deep ResNet architectures were trained using 6521 protein sequences for predicting inter-residue distance and dihedral angles. 208 protein sequences are used to validate the models. The trained neural network is tested using 48 protein sequences whose residue length less than 400 residue from CASP13 dataset are used.

The residue index in the corresponding protein sequence is represented by the x and y-axis of the distance map. The diagonal line shown in each diagram of Fig. 6 conveys that i th– i th residue in protein sequence have a zero distance. Local-range, short-range, medium-range, and long-range distances are classified as $[4.5\text{--}6\text{ Å}^\circ]$, $[6\text{--}12\text{ Å}^\circ]$, $[12\text{--}20\text{ Å}^\circ]$, and all values larger than 20 Å° , respectively.

T 0955 protein sequence is chosen to demonstrate distance prediction done by the proposed method. Figure 6 shows the illustration of distance prediction. The left-hand side of Fig. 6 shows the distance prediction done by the proposed method. The middle one is the distance predicted by AlphaFold. The right-hand side shows the true distance of the query sequence (T0955). By examining the above images the number of pixels value obtained by the proposed method is more than the AlphaFold. It can be concluded that the proposed method produce improved result than the other.

To predict a more accurate protein structure method needs an adequate amount of sequences that exhibit biological similarity with the query sequence. But in some rare cases produce a better result with less no: of sequence alignment. For instance, T0991 has only a few sequence alignments so, neither statistical inferences derived from MSA nor features based on co-evolution support the prediction. The accuracy of the secondary structure is crucial in this scenario.

Figure 7 is the graphical representation of the result obtained by 48 protein sequences in all four methods. The x-axis represents Seq_Id of protein sequences and the y-axis represents TM score. TM score varies from 0 to 1. 0 and 1 stand for low-level and high-level similarity with the corresponding experimental structure respectively. Figure 7a–d shows the bar diagram corresponds to the residues less than 100, 200, 300, and 400 respectively. The red, green, violet, blue bar corresponds to RaptorX, Zhang, AlphaFold, and the proposed method. Examining each figure in Fig. 7 reveals that the proposed method generates a better result than other methods for 25 protein sequences. The table corresponding to Fig. 7 is given in Supplementary File as Table 2.

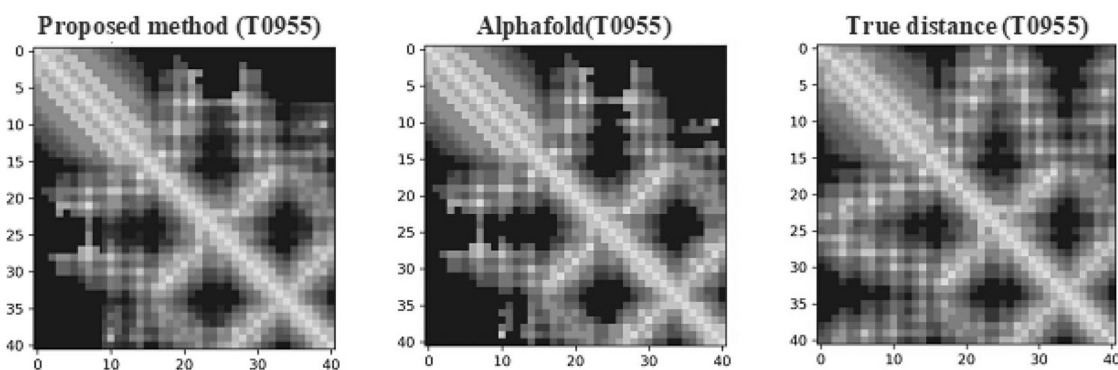


Fig. 6 An illustration of distance prediction

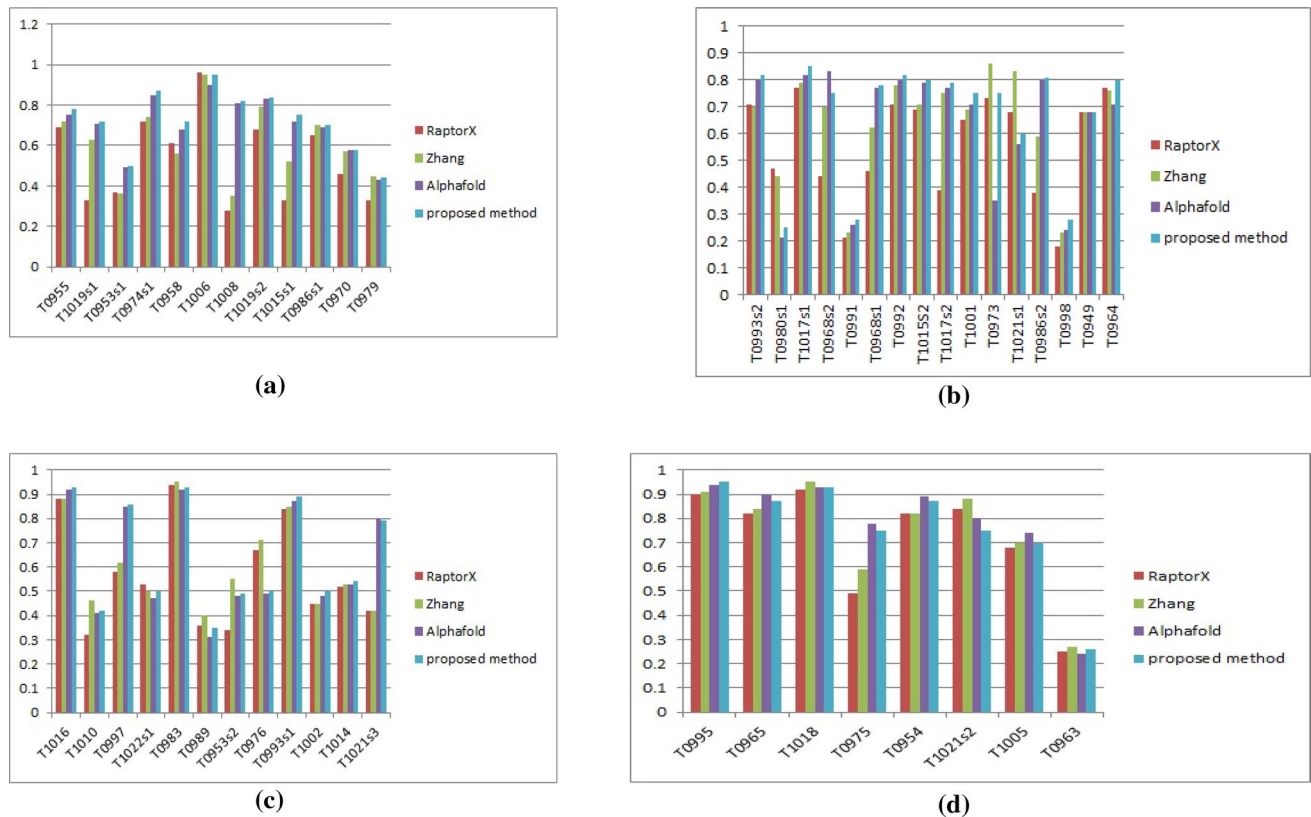


Fig. 7 Individual sequence wise comparison of AlphaFold, Zhang, and RaptorX, the proposed method

Table 1 Comparison of the average TM score

Residue length	Models			
	RaptorX	Zhang	AlphaFold	Proposed method
< 100	0.53	0.61	0.70	0.72
100–200	0.55	0.65	0.62	0.68
200–300	0.57	0.61	0.63	0.64
300–400	0.72	0.75	0.78	0.76
Overall result	0.58	0.65	0.67	0.69

Bold shows the highest TM score in each row corresponding to different residue length

Table 1 shows the comparison of the TM score of AlphaFold, Zhang, and RaptorX, the proposed method based on the range of target length of the CASP13 dataset. From the table, we can see that for the query sequence less than 300 residues the proposed method produces a better result than the other methods. Only for residue length in between 300 and 400 residues, AlphaFold produces better than the proposed method. While considering all 48 protein sequences, it can be concluded the proposed method produces a better result.

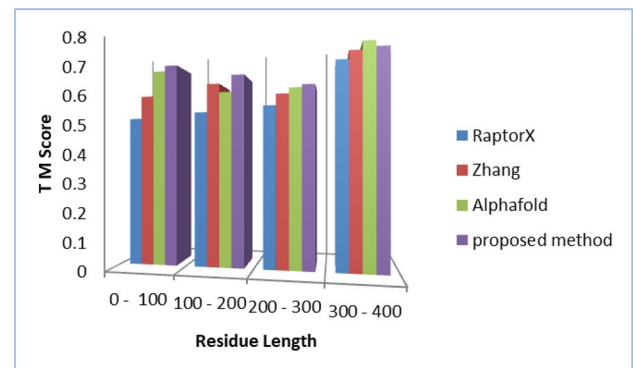


Fig. 8 Comparison of average TM Score obtained by all the methods scaled according to the residue length

Figure 8 visualizes TM score obtained by RaptorX, Zhang, AlphaFold, and the proposed method. The blue, red, green, and violet pillars correspond to the average TM score obtained by RaptorX, Zhang, AlphaFold, and the proposed method based on 0–100, 200–300, 300–400 residue length. From the figure, it can be inferred that the proposed method produces a better TM Score for protein sequences less than 400 residues than others. Figure 8 is the graphical

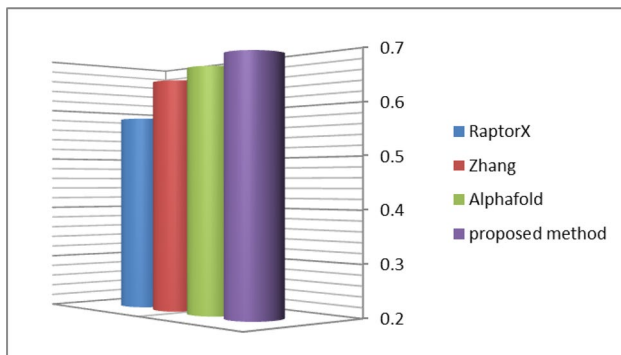


Fig. 9 A visualization of overall TM score of 48 query sequences

representation of Table 1. The x-axis and y-axis in Fig. 8 represent the residue length and TM score respectively.

Figure 9 shows visualize the overall TM score obtained by 48 protein sequences. The blue, red, green, and violet pillars correspond to RaptorX, Zhang, AlphaFold, and proposed method respectively. The figure clearly shows that RaptorX method performs the least among the other methods. While considering protein sequences whose residue length less than 400 residues the proposed method produces a better TM score than other methods. The y-axis in Fig. 9 stands for TM score.

Figure 10 illustrates six 3-D models predicted by the proposed method. Sequence id, number of the amino acid sequence, and TM score obtained are mentioned under each sequence. TM score obtained by each of them is greater than 0.80. While analyzing the result in supplementary file Table 2, it can be inferred that for all six sequences our method produces a better result than others.

The TM score obtained by the proposed method, AlphaFold, Zhang, RaptorX are visualized in Fig. 11. Blue, red, green, violet on Fig. 11 corresponds to RaptorX, AlphaFold, Zhang, and the proposed method respectively. The number on each colour indicates the number of sequences for which that particular method produces a better result than other methods. From the figure we can infer that out of 48 tested sequences, 25 sequences have higher TM score for the proposed method. RaptorX produces the least number of results for the test sequence.

4 Discussion

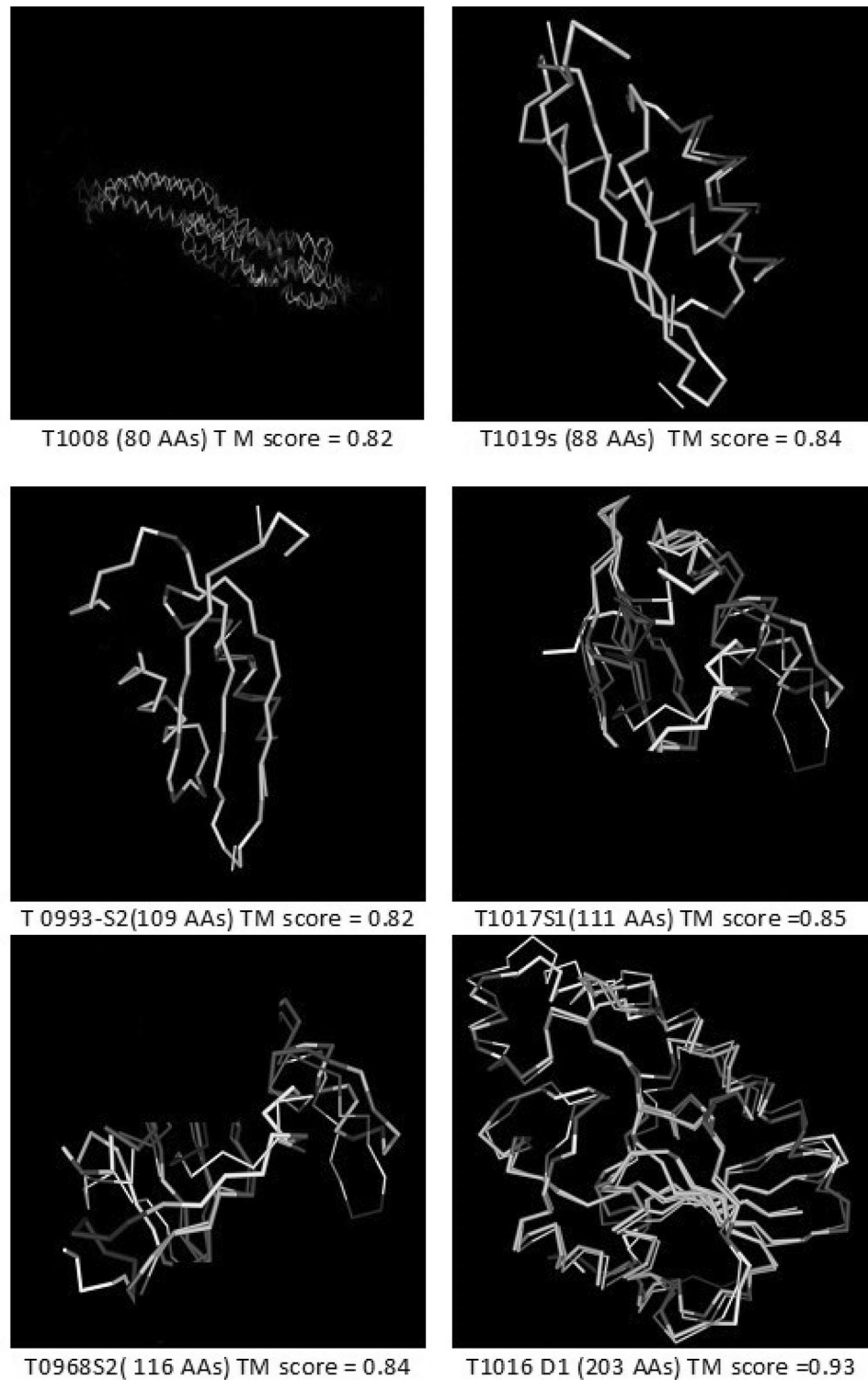
Effective understanding of protein folding helps researchers to develop new drugs, manage biodegradable waste, and unlock enormous benefits. 6521 protein sequences extracted from PDB were used to train the model. For testing 48 target sequences were extracted from CASP 13 dataset whose residue length less than 400 residues are considered.

By applying deep ResNet architecture AlphaFold predicted a 3-D structure of protein while generating inter-residue distance. Later they changed these values to obtain distance potential value. After that sampling technique is used to minimize these values. Finally, the stochastic gradient method is applied to reconstruct the 3-D structure. Zhang also used the ResNet framework to forecast interresidue contact maps and used the fragment-based method to generate 3-d models. While in the case of RaptorX, interatomic distance is predicted using ResNet then transforms to deviation and mean distance to obtain the 3-D model. When we consider contact prediction accuracy both RaptorX and AlphaFold performed better than Zhang but concerning the 3-D formation, Zhang and RaptorX performance were slightly similar. Making use of distance instead of contact estimation has strengthened RaptorX method. In contrast, Zhang used the contact map and a sophisticated folding machine. Since Zhang uses a contact map that contains less information than all the other methods.

Deep neural networks are used in two stages by the proposed method to estimate inter-residue distance and dihedral angles. It incorporates use of varying class width for distance classification. MSA construction process and homologous sequence are generated using a customised sequence database. The use of both evolutionary and non-evolutionary information and gradient algorithm that avoid the need for sampling process altogether contribute 3-D prediction in the proposed method. The accuracy of the predicted protein structure is assessed using the TM score.

While considering 48 test protein sequences, for 52% protein sequences the proposed method produces higher average TM score which are more than AlphaFold, Zhang, and RaptorX. These methods made use of the deep learning approach in the bioinformatics area for three-dimensional PSPs in the CASP13 dataset. There are some cases where the TM score obtained by the proposed method is the same as other methods. Among 25 target sequences, for 10 sequences the proposed method produces a better result than other methods and also has TM Score of more than 0.80. The comparative studies show that the TM scores of AlphaFold,

Fig. 10 List of six 3-D structure of protein predicted by our method



Zhang, and RaptorX are 0.67, 0.65, and 0.58 respectively in the same test set.

Due to the limitation of resources the prediction accuracy became low for long protein sequences. The initial success of the method is motivating. The future aim is to overcome

the limitation of resources to employ more training, and to develop more advanced architecture to predict real-valued distance. And use those to reconstruct accurate 3-D structure prediction.

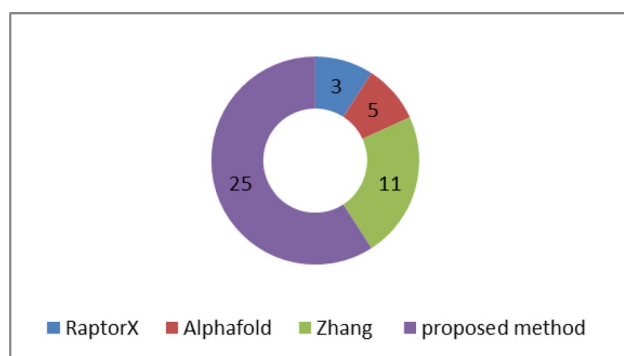


Fig. 11 A visualization of the performance of all methods

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10930-021-10016-7>.

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